Every Value Occurs Infinitely Often

A distribution theorem for the least prime-producing multiplier in OEIS A390117

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Abstract. For the OEIS sequence $a(n)$ given by the least integer $m > 1$ for which $mn + 1$ is prime, we prove that every integer $m \geq 2$ occurs as $a(n)$ for infinitely many positive n. The argument combines finitely many deliberately forced composite values with Dirichlet's theorem on primes in arithmetic progressions. It also proves that the sequence is unbounded, without resolving its conjectured square-root upper bound.

Statement and relation to OEIS

OEIS A390117, introduced by Z.-W. Sun, is defined for $n \geq 1$ by

$$a(n) := \min\{m \in \mathbb{Z} : \ m > 1 \text{ and } mn + 1 \text{ is prime}\}.$$

The minimum exists: Dirichlet's theorem supplies infinitely many primes congruent to $1 \pmod{n}$. The entry conjectures the pointwise upper estimate $a(n) \leq 2\sqrt{n+6}$; this note proves a distinct, unconditional statement about the range and frequency of the values of $a(n)$.

Theorem 1. *For each integer $m \geq 2$, there are infinitely many positive integers n such that*

$$\boxed{a(n) = m.}$$

In fact, these integers n may be selected from a single arithmetic progression on which $kn+1$ is composite for every $2 \leq k < m$, whereas $mn + 1$ is prime for infinitely many n in that progression.

Proof. Fix $m \geq 2$. For every $k \in \{2, 3, \ldots, m-1\}$ choose a different prime $q_k > m$. (If $m = 2$, the collection is empty.) Because $q_k > k$, the residue $-k^{-1}$ exists modulo q_k. The Chinese remainder theorem gives an integer a satisfying the simultaneous conditions

$$ka + 1 \equiv 0 \pmod{q_k} \qquad (2 \leq k < m). \tag{1}$$

Put $Q = \prod_{k=2}^{m-1} q_k$ (with $Q = 1$ when $m = 2$). For every $t \geq 0$, the integer $n = a + tQ$ satisfies all of (1). In particular, for sufficiently large t, each $kn + 1$ with $2 \leq k < m$ is larger than its known proper divisor q_k and is therefore composite.

We now show that the final expression $mn + 1$ is prime for infinitely many of these t. Indeed,

$$mn + 1 = (ma + 1) + mQt.$$

The two coefficients of this progression are relatively prime. Any prime divisor ℓ of m satisfies $ma+1 \equiv 1 \pmod{\ell}$. If instead ℓ divides Q, then $\ell = q_k$ for some $2 \leq k < m$; by (1),

$$ma + 1 \equiv 1 - mk^{-1} \equiv (k - m)k^{-1} \not\equiv 0 \pmod{q_k},$$

because $0 < m - k < q_k$. Thus

$$\gcd(ma + 1, mQ) = 1. \tag{2}$$

By Dirichlet's theorem, the progression $(ma+1)+mQt$ contains infinitely many primes, with t arbitrarily large. For each such t, all earlier candidate values $kn + 1$ $(2 \leq k < m)$ are composite, while $mn + 1$ is prime. Hence the least allowed multiplier is exactly m. □

Corollary 1. *The sequence A390117 is unbounded; moreover, every value in its range occurs infinitely often, and its range is exactly $\{2, 3, 4, \ldots\}$.*

Proof. Immediate from Theorem 1 and the definition $a(n) \geq 2$. □

Example and limitations

For $m = 4$, choose $q_2 = 5$ and $q_3 = 7$. Both constraints $2n + 1 \equiv 0 \pmod 5$ and $3n + 1 \equiv 0 \pmod 7$ hold when $n \equiv 2 \pmod{35}$. On this progression

$$4n + 1 \equiv 9 \pmod{140}, \qquad \gcd(9, 140) = 1.$$

Taking $n = 37$ yields $2n + 1 = 75$, $3n + 1 = 112$, and $4n + 1 = 149$ prime. Thus $a(37) = 4$. The theorem guarantees infinitely many such examples, but does not give an effective bound on the first occurrence of every m, nor does it establish the conjectured inequality $a(n) \le 2\sqrt{n + 6}$.

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References

[1] OEIS Foundation, *A390117: Least integer $m > 1$ such that $mn + 1$ is a prime*, https://oeis.org/A390117, accessed 14 November 2026.

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